

Algorithms Problem Solving

Level 1

Power Of 2 3 4

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Definition

- ◆ This program reads a number from the user and calculates its powers of 2, 3, and 4.
- ◆ It uses the `pow()` function from the `<cmath>` library.
- ◆ For an input of 3, the results are 9, 27, and 81.

Explanation

- ◆ ``ReadNum()`` reads the number from the user and returns it.
- ◆ ``Power_2()`` calculates the number raised to the power of 2.
- ◆ ``Power_3()`` calculates the number raised to the power of 3.
- ◆ ``Power_4()`` calculates the number raised to the power of 4.
- ◆ ``pow(Num, 2)`` means ``Num2```.
- ◆ ``pow(Num, 3)`` means ``Num3```.
- ◆ ``pow(Num, 4)`` means ``Num4```.
- ◆ ``printResult()`` calls the three power functions and prints their results.
- ◆ ``printResult(ReadNum())`` first reads the number and then passes it to ``printResult()``.

Steps in main()

1. Read the number from the user.
2. Calculate the number to the power of 2.
3. Calculate the number to the power of 3.
4. Calculate the number to the power of 4.
5. Print all three results.

💡 Example

◆ If the user enters:

Num = 3

◆ The calculations are:

$$3^2 = 9$$

$$3^3 = 27$$

$$3^4 = 81$$

➡ Result

3 to the Power 2 = 9

3 to the Power 3 = 27

3 to the Power 4 = 81

Note

- ◆ `pow()` is provided by the `<cmath>` library.
- ◆ The function syntax is:
 `pow(base, exponent)`
- ◆ For example:
 `pow(5, 2)` → 25
 `pow(2, 4)` → 16

⊘ Warning

- ◆ `pow()` returns a floating-point value, while these functions return `int`.
- ◆ For the small integer examples used here, the conversion is fine, but very large values can cause precision or integer overflow problems.

Comparison

Power

- ✓ ``pow(Num, 2)`` calculates Num^2 .
- ✓ ``pow(Num, 3)`` calculates Num^3 .
- ✓ ``pow(Num, 4)`` calculates Num^4 .

Multiplication

- ✓ ``Num * Num`` calculates Num^2 .
- ✓ ``Num * Num * Num`` calculates Num^3 .
- ✓ ``Num * Num * Num * Num`` calculates Num^4 .

✦ Summary

- ✓ `pow()` calculates a number raised to a specified power.
- ✓ `pow(Num, 2)` means Num^2 .
- ✓ `pow(Num, 3)` means Num^3 .
- ✓ `pow(Num, 4)` means Num^4 .
- ✓ The program uses separate functions for each power and a function to print the results.

Thank you 😊